Power for Cox Regression

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Introduction

In survival analysis, statistical power is driven by the number of **events**, not by the number of enrolled subjects. A Cox regression power calculation determines the required events and, from assumed accrual and follow-up, the number of subjects.

Prerequisites

Cox proportional hazards, survival analysis.

Theory

For a continuous predictor with standardised-coefficient effect, the required number of events for Wald test power at $1 - \beta$ and two-sided α is

$$D = \frac{(z_{1-\alpha/2} + z_{1-\beta})^2}{\sigma_X^2 \log^2(HR)}.$$

For a binary predictor with allocation π (fraction in one group):

$$D = \frac{(z_{1-\alpha/2} + z_{1-\beta})^2}{\pi(1-\pi) \log^2(HR)}.$$

Required subjects $N = D/p_{\text{event}}$, where p_{event} is the probability of observing the event during follow-up; depends on survival, accrual, and censoring.

Assumptions

- Proportional hazards.
- Independent censoring.
- Known baseline event rate or survival function.

R Implementation

```r
library(powerSurvEpi)

# Two-arm trial: equal allocation, HR = 0.7, 1-year survival = 0.80 in control
ssizeCT.default(power = 0.80, k = 1,
                pE = 1 - 0.70, pC = 1 - 0.80,
                RR = 0.70, alpha = 0.05)

# Number of events for a continuous predictor
# HR per 1-SD increase of 1.4
D <- (qnorm(0.975) + qnorm(0.80))^2 / log(1.4)^2
D
```

Output & Results

For HR = 0.70 with equal allocation and event rates of 20 % (treatment) and 30 % (control), about 500 events and 1000 subjects are required.

Interpretation

"The study requires approximately 500 events to detect a hazard ratio of 0.70 with 80 % power. Assuming the expected event probabilities during follow-up, we need to enrol 1000 patients."

Practical Tips

- Events, not subjects, are the currency of power in survival; count accumulated events, not randomised subjects.
- Report both target events and target N; specify the assumed accrual period and follow-up.
- For interim analyses, use group-sequential boundaries (O'Brien-Fleming, Pocock) to inflate the final sample accordingly.
- Non-proportional hazards reduce power; use restricted mean survival time or weighted log-rank alternatives.
- For competing risks, use Fine-Gray or cause-specific power calculations.

For Reviewers

What to look for in a paper using this method.

- **Common misapplications.**
- **Diagnostics that should be reported but often aren't.**
- **Red flags in tables and figures.**
- **What to verify.**
- **What an adequate Methods paragraph must contain.**

See also — labs in this chapter

- Sample size, power, and Quarto reporting
- Power: closed-form and simulation